not be periodic and the numerical method then computes a discrete sequence of points which gradually fill in the ellipse (see the right panel of Fig. 3.1).

Obviously this conservation property (the existence of a perturbed energy surface) is a special feature of the method we have considered. If we used Euler's method to solve the harmonic oscillator we would find that energy grows without bound.

Now consider the more general Hamiltonian setting. Let $\mathscr{G}_h$ be a rth order symplectic integrator, $r \geq 1$. Suppose that it is the flow map of a certain Hamiltonian system, with Hamiltonian $\tilde{H}_h$. If the method order is r, we may expect this Hamiltonian to be a $\mathcal{O}(h^r)$ approximation of H, thus we posit an expansion of the form

$$\tilde{H}_h = H + h^r H^{(r)} + h^{r+1} H^{(r+1)} + \dots$$

To determine the terms $H^{(r)}, H^{(r+1)}, \dots$, we write the differential equations on $\tilde{H}_h$:

$$\dot{z} = \boldsymbol{J}\nabla H + h^r \boldsymbol{J}\nabla H^{(r)} + h^{r+1}\boldsymbol{J}\nabla H^{(r+1)} + \dots$$

The solution of this system can be expanded in powers of h and equated term-by-term with the expansion of $\mathscr{G}_h$ in powers of h. In this way, successive terms may be computed. Although mechanical, this procedure is tedious. For splitting methods, a more elegant approach is available using the *Lie derivative*, which we next introduce.

3.2 Lie Derivatives and Poisson Brackets

Let $\phi(z)$ be an arbitrary smooth, scalar-valued function of the phase variables (assumed to lie in $\mathbb{R}^m$). At any point $z = \zeta$ in phase space assume that there is a unique solution $z(t, \zeta)$ such that $\dot{z}(t) = \boldsymbol{f}(z(t)), z(0) = \zeta$ that is globally defined for all t. Now differentiate $\hat{\phi}(t) = \phi(z(t))$ with respect to time, using the chain rule, to see how $\phi(z(t))$ is changing as t is varied:

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\hat{\phi}\right|_{t=0} = \left[\sum_{i=1}^{m} \frac{\partial \phi}{\partial z_i}\dot{z}_i\right]\Bigg|_{z=\zeta} = \nabla\phi(\zeta)\cdot\boldsymbol{f}(\zeta).$$

The right hand side suggests a shorthand notation. If we define the Lie derivative $\mathcal{L}_f$ by

$$\mathcal{L}_f\phi = \boldsymbol{f}\cdot\nabla\phi,$$

and note that the origin of time is irrelevant, then the equation for the evolution of ϕ could be written

$$\frac{\mathrm{d}}{\mathrm{d}t}\hat{\phi} = (\mathcal{L}_f\phi)(z). \tag{3.2}$$

Similarly,

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\phi(z(t)) = (\mathcal{L}_f^2\phi)(z(t)).$$

The Taylor series expansion of $\phi(z(t))$ along a solution of the differential equation can therefore be written as

$$\begin{aligned}\phi(z(t)) &= \phi(z(0)) + t\,\frac{\mathrm{d}}{\mathrm{d}t}\phi(z(t))\bigg|_{t=0} + \frac{t^2}{2}\,\frac{\mathrm{d}^2}{\mathrm{d}t^2}\phi(z(t))\bigg|_{t=0} + \dots \\ &= \phi(z(0)) + t(\mathcal{L}_f\phi)(z(0)) + \frac{t^2}{2}(\mathcal{L}_f^2\phi)(z(0)) + \dots \\ &= (e^{t\mathcal{L}_f}\phi)(z(0)).\end{aligned}$$

This gives a concise formula for the evolution of any function of the phase variables, including, in particular, any solution component z_i. Thus the flow map of the system can be represented by $\exp(t\mathcal{L}_f)$. When one writes $\mathscr{F}_t = \exp(t\mathcal{L}_f)$, what is actually meant is that the individual components satisfy

$$z_i(t,\boldsymbol{\zeta}) = [\mathscr{F}_t(\boldsymbol{\zeta})]_i = \left(\exp(t\mathcal{L}_f)z_i\right)\big|_{z=\zeta}.$$

By analogy with the derivation of the matrix exponential, it is tempting to express the exponential as a series expansion in powers of t:

$$e^{t\mathcal{L}_f} = \mathrm{Id} + t\mathcal{L}_f + \frac{t^2}{2}\mathcal{L}_f^2 + \dots$$

However, as $\mathcal{L}_f$ is a first order operator, this series involves high order mixed partial derivatives, and the convergence of the expansion has to be considered in connection with its application to given initial data, that is, we should consider the expansion

$$e^{t\mathcal{L}_f}\phi = \phi + t\mathcal{L}_f\phi + \frac{t^2}{2}\mathcal{L}_f^2\phi + \dots$$

then the primary issue is the boundedness of the terms with respect to functions ϕ of a certain space of functions; we ignore the convergence issue here and treat the operator exponential as a formal series expansion. Methods derived using the formal expansion technique should be subject to further investigation (either numerical or analytical) to establish their validity.